

SKILL – 23

Compilation of Pipeline Monitor Program:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ gcc -Wall -Wextra pipeline_monitor.c -o pipeline_monitor
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ ./pipeline_monitor
=====
OS SKILL 23: PIPELINE MONITOR
=====
Total jobs: 5
Launching jobs...
Launched job 1 with PID 4130
Launched job 2 with PID 4131
Launched job 3 with PID 4132
Launched job 4 with PID 4133
Launched job 5 with PID 4134

Monitoring job completion...
Job 4: simulated failure
PIPELINE-3 PIPELINE RUNNING
PIPELINE-5 PIPELINE RUNNING
PIPELINE-2 PIPELINE RUNNING
PIPELINE-1 PIPELINE RUNNING
Job 1: SUCCESS
Job 2: SUCCESS
Job 3: SUCCESS
Job 4: FAILURE detected
Job 5: SUCCESS

Resource Monitoring:
User CPU time: 0.010229 seconds
System CPU time: 0.006197 seconds
Maximum resident memory: 2268 KB

Performance Analysis:
Elapsed CPU time: 0.008826 seconds

Stability Summary:
Successful jobs: 4
Failed jobs: 1
Total jobs: 5
Pipeline status: FAILURES DETECTED
```

Launching Multiple Pipeline Jobs:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ ./pipeline_monitor
=====
OS SKILL 23: PIPELINE MONITOR
=====
Total jobs: 5
Launching jobs...
Launched job 1 with PID 4149
Launched job 2 with PID 4150
Launched job 3 with PID 4151
Launched job 4 with PID 4152
Launched job 5 with PID 4153

Monitoring job completion...
Job 4: simulated failure
PIPELINE-3 PIPELINE RUNNING
PIPELINE-2 PIPELINE RUNNING
PIPELINE-5 PIPELINE RUNNING
PIPELINE-1 PIPELINE RUNNING
Job 1: SUCCESS
Job 2: SUCCESS
Job 3: SUCCESS
Job 4: FAILURE detected
Job 5: SUCCESS

Resource Monitoring:
User CPU time: 0.014484 seconds
System CPU time: 0.021428 seconds
Maximum resident memory: 2256 KB

Performance Analysis:
Elapsed CPU time: 0.001840 seconds

Stability Summary:
Successful jobs: 4
Failed jobs: 1
Total jobs: 5
Pipeline status: FAILURES DETECTED
```

Resource Monitoring and Performance Analysis:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ ps -u "$USER" -o pid,ppid,stat,comm | head -15
PID  PPID  STAT  COMMAND
379   1      Ss    systemd
380   379    S    (sd-pam)
404   333    S+    bash
4086  4080    Ss    bash
4166  4086    R+    ps
4167  4086    S+    head
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi         551Mi         7.0Gi         3.8Mi         245Mi         7.0Gi
Swap:          2.0Gi           0B           2.0Gi
```

Stability Testing and Failure Detection:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ uptime
06:49:17 up 6:03, 1 user, load average: 0.07, 0.04, 0.00
```

System Architecture and Execution Flow Documentation

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ nano README.md
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-23$ ls -l
total 32
-rw-r--r-- 1 bhardwaj bhardwaj 3249 Oct 10 06:49 README.md
-rw-r--r-- 1 bhardwaj bhardwaj 1 Oct 10 06:48 hello
-rwxr-xr-x 1 bhardwaj bhardwaj 16480 Oct 10 06:48 pipeline_monitor
-rw-r--r-- 1 bhardwaj bhardwaj 3244 Oct 10 06:48 pipeline_monitor.c
```